

Tuza's conjecture for line and total graphs

Samuel Massé

7 September 2026

Abstract

For a graph G , let $\nu_{\Delta}(G)$ be the maximum number of pairwise edge-disjoint triangles and let $\tau_{\Delta}(G)$ be the minimum number of edges meeting every triangle. We prove $\tau_{\Delta}(G) \leq 2\nu_{\Delta}(G)$ when G is the line graph or the total graph of any finite simple graph. The proofs use edge-disjoint incidence cliques and a two-coloring of root edges obtained from Euler tours. A local packing adjustment handles the exceptional Euler-tour seam for line graphs; adjoining each original vertex to the minority color of its incidence clique absorbs the seam for total graphs. The factor two is sharp over line graphs. A sharper orientation cover gives, for every cubic triangle-free graph F , the exact identity $\tau_{\Delta}(\text{Tot}(F)) = \nu_{\Delta}(\text{Tot}(F)) = 5|V(F)|/2$.

1 Introduction

A *triangle packing* in a graph G is a collection of pairwise edge-disjoint triangles, and a *triangle transversal* is a set of edges meeting every triangle. Their maximum and minimum cardinalities are denoted by $\nu_{\Delta}(G)$ and $\tau_{\Delta}(G)$, respectively. Tuza conjectured that

$$\tau_{\Delta}(G) \leq 2\nu_{\Delta}(G) \tag{1}$$

for every finite graph G [5].

We establish (1) for two graph operators built from the incidence structure of a root graph. The *line graph* $L(H)$ has vertex set $E(H)$, with two vertices adjacent when the corresponding root edges meet. The *total graph* $\text{Tot}(F)$ has vertex set $V(F) \cup E(F)$; two elements are adjacent when they are adjacent or incident in F .

Our main results are the following.

Theorem 1.1. *If H is a finite simple graph, then*

$$\tau_{\Delta}(L(H)) \leq 2\nu_{\Delta}(L(H)).$$

The factor two cannot be improved over the class of line graphs.

Theorem 1.2. *If F is a finite simple graph, then*

$$\tau_{\Delta}(\text{Tot}(F)) \leq 2\nu_{\Delta}(\text{Tot}(F)).$$

Corollary 1.3. *If F is a finite simple cubic triangle-free graph, then*

$$\tau_{\Delta}(\text{Tot}(F)) = \nu_{\Delta}(\text{Tot}(F)) = \frac{5}{2}|V(F)|.$$

The two proofs share an edge decomposition. Incidences at a root vertex induce a complete graph, and these complete graphs have pairwise disjoint edge sets. Elementary complete-graph packings give local lower bounds. A single coloring of the root edges controls triangles that cross several incidence cliques. In total graphs we also select all original root edges; the corresponding bridge triangles supply a compatible packing contribution. A balanced orientation improves this cover when the root is triangle-free.

The line-graph inequality for triangle-free roots appears in [2, Corollary 3.3.3]; Theorem 1.1 removes that hypothesis. Nearby results establish Tuza's inequality under a maximum average degree bound [4]; a recent preprint treats maximum degree at most seven [1]. Other results improve the coefficient for certain K_4 -free graphs [3]. Those hypotheses do not cover the unbounded-degree classes considered here. We are not aware of a previous statement of either theorem in the full scope given above.

All graphs in the paper are finite and simple; empty and disconnected roots are allowed. Simplicity ensures that two distinct root edges have at most one common endpoint. The proofs construct an actual transversal C and packing $\mathcal{P}$ in the same graph with $|C| \leq 2|\mathcal{P}|$. No computation or exact formula for complete-graph packing numbers is needed.

2 Complete-graph tools

For $d \geq 0$, put

$$t(d) = \binom{d}{2} - \left\lfloor \frac{d^2}{4} \right\rfloor, \quad D(d) = \nu_{\Delta}(K_d).$$

A balanced bipartition of $V(K_d)$, followed by the selection of every edge inside a part, gives a triangle transversal of size $t(d)$. Explicitly,

$$t(2r) = r(r-1), \quad t(2r+1) = r^2. \quad (2)$$

We use the following elementary lower bounds on $D(d)$. The basic complete-graph inequality is classical; compare the construction attributed to Sebő in [2, Theorem 3.3.2]. We include a proof to make the additional even-order slack explicit.

Lemma 2.1 (Local clique slack). *For every integer $d \geq 0$,*

$$t(d) \leq 2D(d). \quad (3)$$

If $d \geq 6$ is even, then

$$t(d) + 2 \leq 2D(d). \quad (4)$$

Proof. First suppose that $d = 2r$. Split the vertices into sets A and B of size r . Properly edge-color K_A with at most r colors and index the colors by vertices of B . For every edge xy of color b , take the triangle bxy . Each color class is a matching, so the resulting $\binom{r}{2} = t(d)/2$ triangles are edge-disjoint. If $r \geq 3$, all edges inside B remain unused, and we may add one triangle on B . This proves both claims for even d . The required edge coloring follows, for example, from the standard round-robin one-factorization or near-one-factorization of a complete graph.

Now let $d = 2r+1$, with $|A| = r+1$ and $|B| = r$. If r is odd, a one-factorization of the even-order clique K_A has r factors. Indexing them by B and adjoining the index vertex to every factor edge gives $r(r+1)/2$ edge-disjoint triangles. If r is even, use the $r+1$ near-one-factors of the odd-order clique K_A , delete one factor, and index the remaining r factors by B . This gives $r^2/2$ edge-disjoint triangles. In either case twice the number obtained is at least $r^2 = t(d)$. The cases $d \leq 2$ are immediate. $\square$

3 Line graphs

Let H be a finite simple graph. For $v \in V(H)$, let W_v be the set of edges of $L(H)$ corresponding to unordered pairs of distinct root edges incident with v . Two adjacent edges of a simple graph have a unique common endpoint, and hence

$$E(L(H)) = \dot{\bigcup}_{v \in V(H)} W_v. \quad (5)$$

We call W_v the *wedge packet* at v .

Lemma 3.1 (Line-graph triangle classification). *Every triangle of $L(H)$ is exactly one of the following:*

- (i) *a star triangle formed by three root edges incident with one vertex v , whose three edges all belong to W_v ;*
- (ii) *a root triangle formed by the three edges of a triangle xyz in H , with one edge in each of W_x, W_y, W_z .*

Proof. Three distinct pairwise intersecting two-element sets either have a common element or are $\{x, y\}, \{x, z\}, \{y, z\}$. Applied to three pairwise adjacent root edges, these are precisely the two stated cases. $\square$

The star triangles in W_v form the triangle system of $K_{d_H(v)}$. Choose $D(d_H(v))$ edge-disjoint star triangles at every root vertex. The packet partition (5) shows that their union is a packing of size

$$P_0 = \sum_{v \in V(H)} D(d_H(v)). \quad (6)$$

Write

$$T = \sum_{v \in V(H)} t(d_H(v)). \quad (7)$$

Lemma 2.1 gives $T \leq 2P_0$.

The cover uses a balanced red-blue coloring of the root edges. We isolate the required form.

Lemma 3.2 (Alternating Euler coloring). *Let K be a connected graph.*

- (a) *If K has an odd-degree vertex, its edges can be colored red and blue so that the two color degrees are equal at every even-degree vertex and differ by one at every odd-degree vertex.*
- (b) *If every degree of K is even and $|E(K)|$ is even, the two color degrees can be made equal at every vertex.*
- (c) *If every degree of K is even and $|E(K)|$ is odd, the two color degrees can be made equal except at one chosen seam vertex, where they differ by two.*

Proof. For (a), add a new dummy vertex adjacent to every odd-degree vertex. The augmented graph is connected and Eulerian. Start an Euler tour at the dummy vertex and color successive tour edges alternately. At every original vertex the tour edges pair into differently colored entering and leaving edges. Any odd cyclic seam occurs at the dummy vertex. After deleting the dummy edges, the stated balances remain.

For (b) and (c), an isolated vertex needs no coloring. Otherwise alternately color an Euler tour of K . If the tour length is even, every visit pairs two different colors. If it is odd, the first and last edges have the same color and the unique discrepancy of two occurs at the starting vertex. $\square$

Given such a coloring, select in W_v every wedge whose two root edges have the same color. Every star triangle is hit by the pigeonhole principle. Every root triangle is hit because two of its three root edges have the same color. If the red and blue degrees at v are a and b , the local cost is

$$\binom{a}{2} + \binom{b}{2}.$$

By (2), this equals $t(d_H(v))$ when an even degree is balanced or an odd degree differs by one. At a difference-two even degree it equals $t(d_H(v)) + 1$.

Consequently, every connected component has a cover of cost at most its contribution to T , except possibly a connected Eulerian component with an odd number of edges. Such a component has a cover with one extra edge. The following lemma absorbs this unique surcharge.

Lemma 3.3 (Odd Eulerian seam repair). *Let K be a connected Eulerian component of H with an odd number of edges. Put*

$$T_K = \sum_{v \in V(K)} t(d_H(v)), \quad P_K = \sum_{v \in V(K)} D(d_H(v)).$$

There is a triangle transversal C_K of $L(K)$ and a triangle packing $\mathcal{P}_K$ in $L(K)$ such that $|C_K| \leq 2|\mathcal{P}_K|$.

Proof. The alternating Euler cover has size at most $T_K + 1$. We distinguish three exhaustive cases.

If some vertex has degree at least six, its degree is even and (4) supplies two units of local packing slack. Together with (3) at every other vertex, this gives $T_K + 1 \leq 2P_K$.

Suppose next that K is triangle-free. Discard the global coloring. At each vertex, split its incident root edges into two equal parts and select the wedges lying inside a part. Since there are no root triangles, these local selections cover every triangle and have total size exactly T_K . Thus $T_K \leq 2P_K$.

It remains that every positive degree is two or four and that K contains a root triangle xyz . Keep the alternating Euler cover of size at most $T_K + 1$. At a degree-two vertex the local packing is empty. At a degree-four vertex, $D(4) = 1$, and a triangle of the local K_4 can avoid any one prescribed wedge: if that wedge is ab , take the triangle on $\{a, c, d\}$. We may therefore choose the local maximum packings at x, y, z to avoid, independently, the three wedges of the triangle in $L(K)$ corresponding to xyz . Adding it to the local packing gives a packing of size $P_K + 1$. Hence

$$T_K + 1 \leq 2P_K + 1 \leq 2(P_K + 1).$$

These cases are exhaustive because, in the absence of a degree at least six, Eulerianity forces every positive degree to be two or four, after which the component is either triangle-free or contains a root triangle. $\square$

Proof of Theorem 1.1. Apply the alternating Euler cover to every component not requiring the seam repair, and Lemma 3.3 to every connected Eulerian odd-edge component. Line graphs and their triangle systems are disjoint across root components, so the component covers and packings may be united. This proves $\tau_\Delta(L(H)) \leq 2\nu_\Delta(L(H))$.

For sharpness, $L(K_{1,4}) \cong K_4$ has $(\nu_\Delta, \tau_\Delta) = (1, 2)$. Indeed, every two of its triangles share an edge, one edge cannot meet all four triangles, and two opposite edges do. $\square$

Remark 3.4. The quantifier in Lemma 3.3 is one surcharge per connected Eulerian odd-edge component, not per block, root triangle, or high-degree vertex. Odd cycles show why the triangle-free branch must discard the Euler seam rather than charge it to a nonexistent local packing.

4 Total graphs

Let F be a finite simple graph. Write

$$d_v = d_F(v), \quad m = |E(F)|, \quad o = |\{v \in V(F) : d_v \text{ is odd}\}|.$$

For every $v \in V(F)$, put

$$Q_v = \{v\} \cup \delta_F(v),$$

where an edge in $\delta_F(v)$ is regarded as its edge-vertex in the total graph. The set Q_v induces K_{d_v+1} . Its edges are the incidence edges from v to its incident edge-vertices and the wedge edges between pairs of root edges meeting at v . These packet edge sets are pairwise disjoint. Original root edges lie outside all of them. For a root edge uv , let $B_{uv} = \{u, uv, v\}$ be its *bridge triangle*; here uv inside the vertex set denotes the edge-vertex.

Lemma 4.1 (Total-graph triangle classification). *A triangle of $\text{Tot}(F)$ has one of the following forms, according to its number of original vertices:*

- (i) *three original vertices forming a triangle of F ;*
- (ii) *two original vertices and their joining edge-vertex, forming B_{uv} ;*
- (iii) *one original vertex v and two edge-vertices incident with v , all in Q_v ;*
- (iv) *three edge-vertices whose root edges either share an endpoint or form a triangle of F .*

Proof. The first case follows from adjacency of original vertices. In the second, an edge-vertex adjacent to both u and v must be uv . In the third, both edge-vertices must be incident with the original vertex. The fourth follows from the pairwise-intersecting 2-subset argument of Lemma 3.1. $\square$

Proof of Theorem 1.2. Apply Lemma 3.2 componentwise to color the root edges red and blue. At every odd-degree vertex the color degrees differ by one, and at every even-degree vertex they differ by at most two.

In each Q_v , retain these colors on the edge-vertices and color the original vertex v with the less frequent color, breaking ties arbitrarily. Adding one vertex to the minority side changes a discrepancy of zero, one, or two into one, zero, or one, respectively. Thus Q_v is balanced and its within-color edges number exactly $t(d_v + 1)$.

Let C consist of these within-color edges in every Q_v , together with all m original root edges. We check the complete triangle list of Lemma 4.1. Triangles of the first two types have a selected original edge. Those of the third type and the common-endpoint subcase of the fourth lie in a packet and have a same-color pair. In the remaining subcase, two of the three root edges have the same global color, so their joining wedge is selected in their common endpoint's packet. Hence C is a triangle transversal. Since the packet edge sets are disjoint and contain no original root edges,

$$|C| = m + \sum_v t(d_v + 1). \tag{8}$$

For a packing, take all m bridge triangles and, at each v , a packing of $D(d_v)$ triangles among the edge-vertices in $\delta_F(v)$. Distinct bridges use distinct original and incidence edges. The additional triangles use only wedge edges, whose packets are mutually disjoint. Thus their union is one packing $\mathcal{P}$ of size

$$|\mathcal{P}| = m + \sum_v D(d_v) \leq \nu_\Delta(\text{Tot}(F)). \quad (9)$$

The identities

$$t(d+1) - t(d) = \left\lfloor \frac{d}{2} \right\rfloor, \quad \sum_v \left\lfloor \frac{d_v}{2} \right\rfloor = m - \frac{o}{2} \quad (10)$$

follow from (2) and the degree sum. Therefore

$$\tau_\Delta(\text{Tot}(F)) \leq |C| = 2m + \sum_v t(d_v) - \frac{o}{2} \leq 2|\mathcal{P}| - \frac{o}{2} \leq 2\nu_\Delta(\text{Tot}(F)),$$

where the penultimate inequality is Lemma 2.1. $\square$

5 A sharper cover for triangle-free roots

When F is triangle-free, only packet triangles and bridges occur. We can then replace the m selected original edges by a more economical choice.

Proposition 5.1. *For every finite simple triangle-free graph F ,*

$$\tau_\Delta(\text{Tot}(F)) \leq \sum_v t(d_v + 1) + \frac{o}{2} = m + \sum_v t(d_v). \quad (11)$$

Proof. Every graph admits an orientation in which the indegree and outdegree at each vertex differ by at most one: add a dummy vertex adjacent to all odd-degree vertices, orient Euler tours cyclically, and delete the dummy edges. Fix such an orientation of F .

At an even-degree vertex, place v and all incoming edge-vertices in one side of a balanced bipartition of Q_v . Do the same at an odd-degree vertex of indegree $(d_v - 1)/2$. At an odd-degree vertex of indegree $(d_v + 1)/2$, move one incoming edge-vertex to the other side, so the two sides are balanced. Select all edges lying inside a part. This covers every local triangle and uses exactly $t(d_v + 1)$ edges in Q_v .

For an oriented root edge uv directed toward v , the bridge B_{uv} is hit by its incidence edge $v(uv)$ unless uv is the one incoming edge moved at a high-indegree odd vertex. In that exceptional case, select the original edge uv . These additional edges are distinct because every oriented edge has a unique head.

Exactly $o/2$ odd vertices have high indegree: the sum of all indegrees is m , whereas the sum of their lower permitted values is $m - o/2$ by (10). Hence the constructed transversal has $\sum_v t(d_v + 1) + o/2$ edges. The other identity in (10) gives the final equality in (11). $\square$

Proof of Corollary 1.3. Put $n = |V(F)|$. Then $m = 3n/2$ and $D(3) = 1$, so the packing in (9) has $3n/2 + n = 5n/2$ triangles. Proposition 5.1 supplies a transversal of size $m + nt(3) = 3n/2 + n = 5n/2$. Since every transversal contains a distinct edge for each member of an edge-disjoint packing,

$$\frac{5n}{2} \leq \nu_\Delta(\text{Tot}(F)) \leq \tau_\Delta(\text{Tot}(F)) \leq \frac{5n}{2}. \quad \square$$

6 Concluding remarks

The two operator classes permit globally compatible certificates because adjacent root edges have a unique common endpoint. In line graphs, the only possible loss in an Euler coloring is one extra selected wedge per connected Eulerian component with an odd number of edges. In total graphs, the original vertex balances its enlarged incidence clique, while bridge triangles pay for the selection of original root edges.

The factor two is sharp for line graphs. We do not determine the best factor for total graphs, although the cubic triangle-free case has equality between packing and transversal numbers. Extensions to multigraph roots, generalized line graphs, or quasi-line graphs require additional arguments and are not claimed here.

AI assistance. The author acknowledges the assistance of OpenAI’s GPT-6 Astra in developing the proofs, checking arguments, searching the literature, and preparing the exposition. The proofs are self-contained; computational checks are not premises of the theorems.

References

- [1] Anish Gupta. Tuza’s conjecture for graphs of maximum degree at most seven, 2026. Preprint, arXiv:2608.06538v1.
- [2] Andrea Munaro. *Sur quelques invariants classiques et nouveaux des hypergraphes*. PhD thesis, Université Grenoble Alpes, 2016.
<https://andreamunaro.github.io/munaro-thesis.pdf>.
- [3] Andrea Munaro. Triangle packings and transversals of some K_4 -free graphs. *Graphs and Combinatorics*, 34(4):647–668, 2018. doi:10.1007/s00373-018-1903-y.
- [4] Gregory J. Puleo. Tuza’s conjecture for graphs with maximum average degree less than 7. *European Journal of Combinatorics*, 49:134–152, 2015.
doi:10.1016/j.ejc.2015.03.006.
- [5] Zsolt Tuza. A conjecture on triangles of graphs. *Graphs and Combinatorics*, 6(4):373–380, 1990. doi:10.1007/BF01787705.